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#include <stdio.h>
#include <stdlib.h>
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/******
```

function name: convertToDifferentBase

description: Convert a decimal number @dec to the number with different @base and store the result as a string in result

return vaule: Result after convert.

notice: The letter you see in the number should be upper case.

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*****/
```

```
char *convert(int dec, int base, char *result)
```

```
{
    char hex[] = {'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F'};

    if (dec == 0)
    {
        return result;
    }

    int temp = dec % base;
    int size = strlen(result);

    char ans[size+2];
    ans[0] = hex[temp];
    ans[1] = '\0';

    strcat(ans, result);
    strcpy(result, ans);

    return convert(dec / base, base, result);
}
```

```
char *convertToDifferentBase(int dec, int base, char *result)
```

```
{
    result[0] = '\0';
    return convert(dec, base, result);
}
```